

Phase 1 – Dataset & Research Proposal

Team Members:

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“Comparing the Impact of Covid-19 Policies on Mental Health across U.S. States”

Research Question

How heavily did a state’s COVID-19 measurement policy have an impact on mental health compared to its counterparts?

We aim to find a way to statistically compare COVID-19 policies of different U.S. states and what impact they had on the mental health rates of their population, based on the policy’s strictness, and measured depression and anxiety changes in the populations. In nature, this project is both descriptive and explanatory, because we will describe how anxiety/depression rates change across states over time and test whether states with stricter policies show significantly different mental health trends than their counterparts. One row in our dataset represents a survey time period for a specific state and population subgroup, including the estimated percentage reporting anxiety and depressive symptoms, along with its confidence interval.

Dataset Information

- **Dataset Source Link:**
<https://catalog.data.gov/dataset/indicators-of-anxiety-or-depression-based-on-reported-frequency-of-symptoms-during-last-7->
- **Dataset Download Link:**
<https://data.cdc.gov/api/views/8pt5-q6wp/rows.csv?accessType=DOWNLOAD>
- **Data Collection:** Household Surveys
- **License:** Not subject to copyright (public)
- **Number of rows:** 16,795
- **Number of columns:** 14
- **Data types:** numeric, categorical, text, date
- **Time Range:** April 2020 until September 2024
- **Known limitation:** The dataset was put quickly into the field, could be subject to sampling error

Planned EDA Questions

- Are there missing values in our key variables (time period, value, confidence)
- Are the individual survey samples comparable, or do we need to make adjustments to make them more representative?
- Can we find certain time rates where we can observe a huge change in the survey results across many to all U.S. states?
- Can we detect sampling errors, by comparing the low CI, high CI, and quartile range?
- Do the different states show a similar increase or decrease in their measured anxiety/depression rates, or do they vary a lot?
- Are there major variances between the results of different population sub-groups (such as age, sex, ethnicity)?

Anticipated Challenges

- Complexity arising from including value, low/high CI, and quartile range in our calculations or analysis
- Differences in data sampling between phase 1 to 3.1, compared to phase 3.2 and beyond might impact the survey results (*the question in the housing survey was replaced with a different question*)
- Policy Variance is not included: Dataset doesn't contain "lockdown strictness" so a separate dataset will be required possibly
- The Policy Variance needs to be numerically comparable, especially in terms of giving a statistically way to express "how strict" they were
- The policies of different states changed over time, resulting in variances in the survey results. This might lead to very different results for our research question in different time frames during the Covid lockdown